

PROJECT REPORT

Data Visualization

DecodeLabs Data Analytics Internship
Optional Mastery Phase | Batch 2026

Project Title	Data Visualization
Organization	DecodeLabs
Program	Data Analytics Internship – Industrial Training Kit
Batch	2026
Tool Used	Python + Matplotlib / Seaborn
Date	15 September 2026

1. Introduction

This report documents the Data Visualization milestone. The goal was to move from finding patterns to explaining them — producing clear, honest, boardroom-ready charts that communicate insights within five seconds.

Visualization is not decoration. It is the final translation layer between analytical work and executive action. Every chart in this project follows three pillars: choose with purpose, edit with precision, and tell a story.

2. Project Objectives

- Create appropriate chart types (bar, line, scatter) matched to the analytical question.
- Apply data-ink ratio principles and eliminate chartjunk.
- Use action titles that state the conclusion, not just the topic.
- Highlight key insights with color as a spotlight.
- Answer the “So What?” question for each visual.

3. Design Principles Applied

- Horizontal bar charts for categorical comparisons (better label readability).
- Line chart for time trends.
- Scatter plot for relationship exploration.
- Zero baseline on all bar charts (axis integrity).
- Direct data labels instead of separate legends where practical.
- Single accent color to highlight the key category; muted greys for context.

4. Key Visual Insights

Chart	Action Insight
Revenue by Product	Chair and Printer lead revenue — nearly equal top performers
Order Status	Cancellation + Return = 41% of all orders — major operational risk
Monthly Trend	Order volume remains relatively stable across 30 months
AOV by Product	Laptop has the highest average order value at \$1,111
Referral Source	Instagram drives the highest revenue among referral sources
Payment Method	Online payments contribute the largest share of revenue

5. Business Recommendation

The strongest visual signal is the combined 41% cancellation and return rate. This should be the priority investigation area. Secondary opportunities include protecting Instagram as the top acquisition channel and exploring why Laptop commands the highest average order value.

6. Deliverables

- `cleaned_dataset.xlsx` – Source dataset
- `data_visualization.py` – Reproducible visualization script
- `viz_outputs/` – Eight insight-driven PNG charts
- `Project_Report_Visualization.pdf` – This formal report
- `README.md` – Project documentation

7. Conclusion

This optional mastery phase demonstrates the ability to translate analytical findings into clear visual narratives. By applying chart selection discipline, data-ink principles, and action-oriented titles, the visuals support faster and more confident decision-making.

Data visualization is the bridge between insight and action. This project completes that bridge.